

Topological Memory.

David Ponarovsky

November 24, 2025

Introduction

Today:

- ▶ Proof.
 1. Walkthrough the math

Our Code

Code

In this session, we consider LDPC codes, where each bit adjoins to Δ_1 checks, and each check adjoins to Δ_2 bits. We denote by $\Delta = \Delta_1 \Delta_2$.

Local Decoder

definition

We say that D is a local decoder if when applied against $2p(1-p)$ -depolarized channel:

- ▶ There exists function $M : (0, 1) \rightarrow (0, 1)$ such that the probability of each bit being flipped is $M(2p(1-p)) < p^{\frac{100}{99}}$.
- ▶ There is a constant l such that for any bit, there are only $(\Delta)^l$ bits on which the event of it being flipped depends.

For example, if we take k large enough and consider the (n, k) -Toric, then both the swift-rule and the flip upon majority are local decoders.

Storing Logical State - Classical Statement

Logical State

For a random variable X we denote by:

1. $\mathbf{E}_{\sim p}[X]$ - its expectation when measured after a single application of the p -depolarizing channel.
2. $\mathbf{E}_{\sim q^{(t)}}[X]$ - its expectation when measured after t rounds, each initiated by a p -depolarized channel followed by an application of local decoder D .

The Theorem - Classical Statement

Theorem

There is $p_{\text{th}} \in (0, 1)$ such that for any $p < p_{\text{th}}$ and any bounded function f with a supremum $|f|$ it holds:

$$\mathbf{E}_{\sim q(t)} [f] \leq \mathbf{E}_{\sim p} [f] + t|f|e^{-\Theta(\sqrt{n})}$$

Lemma

There exists a constant $c > 0$, such:

$$\mathbf{E}_{\sim 2p(1-p)} [f \circ D] \leq \mathbf{E}_{\sim p} [f] + |f|e^{-c\sqrt{n}}$$

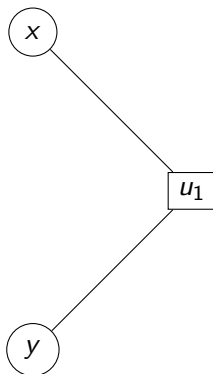
Theorem Proof

By induction on t .

$$\begin{aligned}\mathbf{E}_{\sim q^{(t)}}[f] &= \mathbf{E}_{\sim q^{(t-1)}}[\mathbf{E}_{\sim p}[f \circ D]] \\ &\leq \mathbf{E}_{\sim p}[\mathbf{E}_{\sim p}[f \circ D]] + (t-1)|\mathbf{E}_p[f \circ D]|e^{-c\sqrt{n}} \\ &\leq \mathbf{E}_{\sim 2p(1-p)}[f \circ D] + (t-1)|f|e^{-c\sqrt{n}} \\ &\leq \mathbf{E}_{\sim p}[f] + t|f|e^{-c\sqrt{n}}\end{aligned}$$

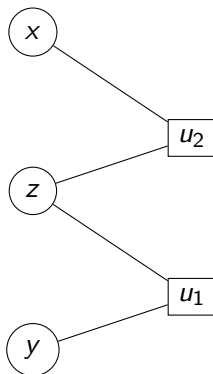
When in the last transition we used the lemma.

Dependence Type I



$$\begin{aligned} & \mathbf{Pr}[(x, y) \in e | u_1 \text{ is satisfied}] \\ &= \mathbf{Pr}[x \in e | u_1 \text{ is satisfied}] \mathbf{Pr}[y \in e | u_1 \text{ is satisfied}] \\ &\leq \mathbf{Pr}[x \in e] \mathbf{Pr}[y \in e] \end{aligned}$$

Dependence Type II



$$\begin{aligned} & \mathbf{Pr}[(x, y) \in e | z \text{ is not flipped}] \\ &= \mathbf{Pr}[x \in e | z \text{ is not flipped}] \mathbf{Pr}[y \in e | z \text{ is not flipped}] \\ &\leq \mathbf{Pr}[x \in e] \mathbf{Pr}[y \in e] \end{aligned}$$

Definition

Consider the moment after the application of the depolarizing channel, and right before the correction by the decoder.

For a subset of qubits S , we say that a qubit $x \in S$ is depended if there exists $y \in S$ such:

1. Either x and y share a check that is not satisfied. (Type I)
2. Or, x and y have checks which share a bit that was flipped by the application of the channel. (Type II)

Dependence

Claim

Consider the application of a $2p(1 - p)$ -depolarizing channel. Let S be a subset of qubits chosen uniformly random¹, then the probability that there are more than $\frac{1}{100}$ depended qubits in S is less than $2e^{-c\sqrt{n}}$.

¹each qubit is taken with probability $\frac{1}{2}$

Dependence

Proof

First, for an arbitrary S , denote by J the qubits that depend on qubits in S . Then, the expected number of depended qubits in S is:

$$\begin{aligned} & \mathbf{E}_{\sim 2p(1-p)} [\text{Type II depended qubits in } S] \\ & \leq \mathbf{E}_{\sim 2p(1-p)} [|J|] \leq 2p(1-p)\Delta' |S| \end{aligned}$$

Similarly, let J' be the qubits in S that connected to unsatisfied check, then:

$$\begin{aligned} & \mathbf{E}_{\sim 2p(1-p)} [\text{Type I depended qubits in } S] \\ & \leq \mathbf{E}_{\sim 2p(1-p)} [|J'|] \leq \frac{1 - (1 - 4p(1-p))^{\Delta_2}}{2} \Delta' |S| \end{aligned}$$

Dependence

We are going to use the following inequality:

Theorem 1 (McDiarmid's inequality). *Let $X_1, \dots, X_n$ be independent random variables, where X_i has range $\mathcal{X}_i$. Let $f: \mathcal{X}_1 \times \dots \times \mathcal{X}_n \rightarrow \mathbb{R}$ be any function with the $(c_1, \dots, c_n)$ -bounded differences property: for every $i = 1, \dots, n$ and every $(x_1, \dots, x_n), (x'_1, \dots, x'_n) \in \mathcal{X}_1 \times \dots \times \mathcal{X}_n$ that differ only in the i -th coordinate ($x_j = x'_j$ for all $j \neq i$), we have*

$$|f(x_1, \dots, x_n) - f(x'_1, \dots, x'_n)| \leq c_i.$$

For any $t > 0$,

$$\Pr(f(X_1, \dots, X_n) - \mathbb{E}[f(X_1, \dots, X_n)] \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right).$$

Dependence

Thus, the probability that the number of depended qubits in S is $|S|^{\frac{3}{4}}$ far from the expectation:

$$\Pr_{\sim 2p(1-p)} \left[\# - \mathbf{E}[\#] \geq |S|^{\frac{3}{4}} \right] \leq e^{-2 \frac{|S|^{\frac{3}{2}}}{n \Delta^I}}$$

So, for any $|S| > \alpha n$ we have:

$$\Pr_{\sim 2p(1-p)} \left[\# - \mathbf{E}[\#] \geq |S|^{\frac{3}{4}} \right] \leq e^{-\frac{2\alpha^{\frac{3}{2}}}{\Delta^I} \sqrt{n}}$$

Dependence

Thus for small enough $p \in (0, 1)$, with probability greater than $1 - 2e^{-\frac{2\alpha^{\frac{2}{3}}}{\Delta'}\sqrt{n}}$, the number of depended bits is less than:

$$\left(2p(1-p) + \frac{1 - (1 - 4p(1-p))^{\Delta_2}}{2}\right) \Delta' |S| + |S|^{\frac{3}{4}} \rightarrow \frac{|S|}{100}$$

Lemma Proof

Denote by S the subset of the flipped qubits, and denote by $de(S)$ the number of depended qubits in S .

$$\begin{aligned}\mathbf{E}_{\sim 2p(1-p)}[f \circ D] &= \sum f \circ D(S) \cdot \mathbf{Pr}[S] \\ &\leq \sum_{\substack{de(S) < \frac{1}{100}|S| \\ |S| \geq \alpha n}} f \circ D(S) \cdot \mathbf{Pr}[S] + 2|f|e^{-c\sqrt{n}} \\ &\leq \sum_{\substack{de(S) < \frac{1}{100}|S| \\ |S| \geq \alpha n}} f(S)M(2p(1-p))^{\frac{99}{100}|S|} + 2|f|e^{-c\sqrt{n}}\end{aligned}$$

Lemma Proof

$$\begin{aligned} &\leq \sum_{\substack{de(S) < \frac{1}{100}|S| \\ |S| \geq \alpha n}} f(S)p^{|S|} + 2|f|e^{-c\sqrt{n}} \\ &\leq \sum_S f(S)p^{|S|} + 2|f|e^{-c\sqrt{n}} \\ &\leq \mathbf{E}_{\sim p}[f(S)] + 2|f|e^{-c\sqrt{n}} \end{aligned}$$

Storing Logical State - Quantum Statement

Logical State

Let $\rho^{(0)}$ be a mixed state over codewords. Define:

$$\rho^{(t)} = D \circ \mathcal{N} \rho^{(t-1)}$$

1. $\mathbf{E}_{\sim p}[X]$ - its expectation when measured after a single application of the p -depolarizing channel.
2. $\mathbf{E}_{\sim q^{(t)}}[X]$ - its expectation when measured after t rounds, each initiated by a p -depolarized channel followed by an application of local decoder D .

The Theorem - Quantum Statement

Theorem

There is $p_{\text{th}} \in (0, 1)$ such that for any $p < p_{\text{th}}$ and any bounded Hermitian operator X we have:

$$\mathbf{Tr} X\rho^{(t)} \leq \mathbf{Tr} X\mathcal{N}_p(\rho) + t\|X\|e^{-\Theta(\sqrt{n})}$$

Lemma

There exists a constant $c > 0$, such:

$$\mathbf{Tr} XD \circ \mathcal{N}_p \circ \mathcal{N}_p(\rho) \leq \mathbf{Tr} X\mathcal{N}_p(\rho) + 2\|X\|e^{-\Theta(\sqrt{n})}$$

Theorem Proof

By induction on t .

$$\begin{aligned}\mathbf{Tr} X\rho^{(t)} &= \mathbf{Tr} XD \circ \mathcal{N}_p \rho^{(t-1)} \\ &\leq \mathbf{Tr} XD \circ \mathcal{N}_p \circ \mathcal{N}_p(\rho) + (t-1)\|X\|e^{-\Theta(\sqrt{n})} \\ &\leq \mathbf{Tr} X\mathcal{N}_p(\rho) + t\|X\|e^{-\Theta(\sqrt{n})}\end{aligned}$$

When in the last transition we used the lemma.

Corollary

$$\begin{aligned}\mathbf{Tr} (I - X) \rho^{(t)} &\leq \mathbf{Tr} (I - X) \mathcal{N}_p(\rho) + t \|I - X\| e^{-\Theta(\sqrt{n})} \\ \Leftrightarrow \mathbf{Tr} X \rho^{(t)} &\geq \mathbf{Tr} X \mathcal{N}_p(\rho) - t \|I - X\| e^{-\Theta(\sqrt{n})}\end{aligned}$$

Corollary II

$$\begin{aligned} & \mathbf{Tr} \, XID\mathcal{N}_pID\mathcal{N}_pID\mathcal{N}_p..ID\mathcal{N}_p\hat{U}D\mathcal{N}_pID\mathcal{N}_p..ID\mathcal{N}_p\rho \\ & \leq \mathbf{Tr} \, XID\mathcal{N}_pID\mathcal{N}_pID\mathcal{N}_p..ID\mathcal{N}_p\hat{U}\mathcal{N}_p\rho + e^{-t_1} \end{aligned}$$